
IC 115: ODE and Integral transform

Course Code	IC 115
Course Name	ODE and Integral transform
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	B.Tech 1st year
Mutual Exclusion	None
Approval	50th BoA (Jul 2023); 44th Senate (Sep 2024), in the B.Tech (all branches) - Institute Core curriculum

Course Contents

- **Ordinary Differential Equations:** Origin of differential equations, Formation of differential equations, Order and degree, Equation of first order and first degree, Solution of linear differential equations with constant coefficients, Euler Cauchy Equations, Solution of Second Order differential Equations by change of dependent and independent variables, Method of variation of parameters for second order differential equations, Series solution. [13 Lectures]
- **Integral transforms:** Laplace and Fourier transform, existence, linearity property, shifting property, Inverse Laplace and Fourier, Melin transform, Fourier series [8 Lecture]

Text Books

1. G. F. Simmons, Ordinary Differential Equations, Differential equations with applications and historical notes, 2nd Edition.
2. S. L. Ross, Introduction to Ordinary Differential Equations, Wiley, 1980.

References

1. Gilbert Strang, Introduction to Applied Mathematics.
2. E. Kreyszig, Advanced Engineering Mathematics